

Accurate and Efficient Long-Term Memory for LLM Agents

Zicheng Zhao^{1,†}, Xinyang Guo^{1,†}, Luyao Lv², Menghan Wang²,
Ming Li^{1,*}, Shuaicheng Li^{3,*}

¹Central China Research Institute for AI Technology

²Guangdong Provincial Key Laboratory of Interdisciplinary Research
and Application for Data Science, BNU-HKBU United International College, Zhuhai

³Department of Computer Science, City University of Hong Kong

[†]Equal contribution ^{*}Corresponding authors

Abstract

LLM agents augmented with persistent memory can recall past interactions, but existing systems suffer from two limitations: flat, unstructured storage loses relational context needed for multi-hop and temporal reasoning, and reliance on expensive LLM-based classification makes them impractical for latency-sensitive deployment. Without mechanisms to validate new information against stored knowledge, these systems silently accumulate contradictions. We present MOSAIC (Memory-Organized Structured Agent for Information Collection), a structured, conflict-aware long-term memory framework for LLM agents that is substantially more accurate and efficient. MOSAIC introduces three key capabilities: (1) entity-typed graph storage with semantic classification preserving relational structure across events, personas, and relationships, enabling multi-hop and temporal reasoning over conversation history; (2) hash-accelerated dual-path retrieval replacing LLM-based classification with locality-sensitive hashing, achieving near-instantaneous lookup with negligible accuracy loss; and (3) active conflict detection at save time that cross-references new information against existing graph neighbors, triggering updates or deletions for contradictory entries. Evaluated on LoCoMo (long-conversation QA), HaluMem, and a novel clinical-guideline error compounding test, MOSAIC achieves 89.35% accuracy on LoCoMo (+27.21 pp over the best baseline), best HaluMem-Medium extraction F1 (86.77%) and HaluMem-Long extraction F1 (85.84%), best QA correctness on both Medium and Long (73.10%, 70.75%), and detects 66% of injected factual conflicts-4.7 times higher than the best baseline (14%)-while hash-accelerated retrieval keeps average search latency at 0.58 s per question.

Keywords: LLM agents, long-term memory, memory graph, conflict detection, error compounding, conversational QA

1 Introduction

The proliferation of memory-augmented LLM agents has created an implicit assumption: that an agent which remembers is an agent that reasons correctly over its memories. Systems such as MemGPT [Packer et al., 2023], Mem0 [Chhikara et al., 2025], A-Mem [Xu et al., 2025], and MemOS [Li et al., 2025] have demonstrated impressive retention of user preferences, factual history, and contextual nuance across sessions. Yet a fundamental limitation persists: these systems treat memory as *append-only storage*, ingesting new information without validating it against existing knowledge, and thus silently accumulating contradictions that compound across retrieval and reasoning stages.

This *error compounding* problem is particularly acute in safety-critical domains. In clinical settings, silently storing that a patient’s target blood pressure is <160/100 mmHg when guidelines specify <130/80 mmHg propagates into incorrect risk stratification and potentially harmful prescribing [Yaraghi, 2024]. In legal and financial contexts, contradictory facts feed contradictory retrievals, which feed contradictory downstream decisions. The pipeline structure of modern memory systems—from extraction to storage to retrieval to reasoning—means that a single undetected error at ingestion can exponentially amplify through subsequent stages.

A growing body of evidence reveals a deeper disconnect between *recall* and *task completion*: memory-augmented agents routinely terminate conversations prematurely, miss required information, or drift into tangential topics when no explicit mechanism enforces systematic coverage [Xi et al., 2025, Wang et al., 2024]. Current agent frameworks provide two capabilities: *action selection* (choosing what to do next) and *memory* (storing what was learned). But structured information-gathering tasks require a third: *task-state tracking*—a persistent, queryable representation of which required information units have been obtained, which remain outstanding, and what dependencies constrain the order in which they should be

sought. Without this, an agent cannot distinguish between “I have enough information to stop” and “I have run out of things that are easy to ask.”

This distinction is not merely an engineering detail. In clinical intake, a missed medication history propagates into incorrect risk stratification and potentially harmful prescribing [Singhal et al., 2023]. In insurance claims processing, a missing liability determination invalidates downstream adjudication. In technical support, an undiagnosed hardware failure leads to repeated software-only fixes. The common structure across these domains is a *partially observable information space* with *prerequisite constraints* and *coverage requirements* that must be systematically satisfied through multi-turn interaction.

We trace these failures to three structural deficiencies in existing memory architectures: *flat storage*, where memories are kept as unstructured text chunks or key-value pairs without typed relationships, making cross-referencing impossible; *passive ingestion*, where new information is stored unconditionally without conflict checking against existing memories; and *monolithic retrieval*, where semantic search over flat stores cannot exploit relational structure for multi-hop or temporal reasoning.

A deeper question is *why graphs* rather than simpler task representations such as flat checklists, priority queues, or plan templates. The answer lies in a structural property of real information-gathering tasks: *local dependency*. Whether a particular information entity needs to be queried, re-queried, or deprioritized depends not on the global state of the entire conversation but on the states of a small set of semantically or logically related entities—its *graph neighbors*. Knowing a patient’s creatinine level affects the importance of asking about nephropathy but not about smoking history. Confirming the date of an accident changes the relevance of weather-condition queries but not vehicle-make queries. This locality is precisely what graph structure encodes and what flat representations discard.

We introduce **MOSAIC**, a framework that addresses these deficiencies through three mechanisms. First, MOSAIC organizes memories as a *typed entity graph* with semantic classification into events, personas, and relationships, preserving relational structure that enables graph-based reasoning. Second, MOSAIC performs *active conflict detection at save time*: when new information is ingested, it is cross-referenced against the existing memory graph; conflicts trigger warnings and may be resolved through updates or deletions. Third, MOSAIC employs *hash-accelerated retrieval* that achieves near-instantaneous memory lookup while maintaining accuracy comparable to LLM-based classification.

We formalize the key insight underlying MOSAIC’s graph structure as *neighbor-conditioned stability* (NCS): a node’s importance score, belief state, and priority require re-evaluation only when at least one of its graph neighbors has changed state. When the local neighborhood is

unchanged, the node’s state is guaranteed stable. This principle is closely analogous to Bellman’s principle of optimality in dynamic programming [Bellman, 1966], which decomposes a global optimization into local value updates that propagate through a state-transition graph. Just as dynamic programming avoids redundant global recomputation by exploiting local structure, MOSAIC avoids redundant re-evaluation of entity priorities by exploiting the dependency and association structure encoded in the memory graph.

We evaluate MOSAIC on three complementary benchmarks: LoCoMo [Maharana et al., 2024], which tests long-conversation memory QA across 1,540 question-answer pairs; HaluMem [Chen et al., 2025], which separately evaluates memory extraction, updating, and question answering with hallucination detection; and a novel error compounding test in which 50 factual conflicts are manually injected into hypertension clinical guidelines. Across these benchmarks, MOSAIC achieves 89.35% overall accuracy on LoCoMo (+27.21 pp over the best baseline Mem0), best HaluMem-Medium extraction F1 (86.77%), best HaluMem-Long extraction F1 (85.84%), best HaluMem QA correctness on both Medium and Long (73.10%, 70.75%), detects 66% of injected factual conflicts— $4.7\times$ higher than the best baseline (14%)—and keeps average HaluMem-Medium search latency at 0.58 s per question.

2 Related Work

2.1 Memory-Augmented LLM Agents

The development of persistent memory for LLM agents has progressed rapidly. MemGPT [Packer et al., 2023] introduces OS-inspired virtual memory management, maintaining a tiered memory hierarchy (main context, archival storage, recall storage) with explicit memory management functions. Mem0 [Chhikara et al., 2025] provides multi-level persistent memory with adaptive personalization across user, session, and agent levels. A-Mem [Xu et al., 2025] implements adaptive memory with automatic storage, retrieval, and summarization, dynamically adjusting memory granularity. MemOS [Li et al., 2025] unifies memory operations through an operating-system abstraction layer. Zep [Rasmussen et al., 2025] provides production-grade memory with temporal awareness for AI assistants.

A comprehensive survey by Zhang et al. [2025] categorizes memory mechanisms along dimensions of storage format, retrieval strategy, and management policy. Despite this diversity, all existing systems share a common limitation: they focus on *retention*—ensuring that information, once stored, can be retrieved accurately—but lack mechanisms for *conflict detection* or *structured organization* that would prevent contradictory information from being stored in the first place.

2.2 Agent Frameworks and Planning

The broader landscape of LLM agent architectures provides context for MOSAIC’s design choices. ReAct [Yao et al., 2022] interleaves reasoning and acting, using chain-of-thought prompting to decide actions step by step. Reflexion [Shinn et al., 2023] adds verbal reinforcement learning through self-reflective feedback loops. Plan-and-Solve [Wang et al., 2023] improves zero-shot reasoning through explicit plan generation before execution. These frameworks demonstrate sophisticated planning and reasoning capabilities but lack persistent, structured memory for cross-session information management [Xi et al., 2025, Wang et al., 2024].

Tool-augmented approaches [Schick et al., 2023] extend agents with external capabilities but do not address the fundamental memory organization problem. Chain-of-thought prompting [Wei et al., 2022] improves reasoning within a single context window but does not support long-term memory across sessions.

2.3 Knowledge Graphs and Structured Memory

Several systems organize agent memory as knowledge graphs [Pan et al., 2024]. Mem0-Graph extends Mem0 with graph structure but does not perform conflict checking. GraphRAG [Edge et al., 2024] uses graph-based retrieval for query-focused summarization, demonstrating the benefit of graph structure for information retrieval. However, these systems focus on retrieval quality rather than ingestion-time validation.

MOSAIC’s entity graph differs from general knowledge graphs in its focus on typed entities (event, persona, relationship) with explicit conflict detection at ingestion time. The dual-graph formulation—separating prerequisite dependencies from semantic associations—provides both logical validity and conversational coherence, properties that single-graph approaches cannot simultaneously optimize.

2.4 Memory Evaluation Benchmarks

LoCoMo [Maharana et al., 2024] evaluates long-conversation memory through multi-type QA spanning single-hop, multi-hop, temporal, and open-domain questions. HaluMem [Chen et al., 2025] introduces hallucination-aware evaluation of the memory pipeline, separately assessing extraction, updating, and question answering stages. LLM-as-Judge approaches [Zheng et al., 2023] supplement reference-based evaluation. Our error compounding test complements these by specifically targeting conflict detection, a capability not evaluated by existing benchmarks.

2.5 Hallucination and Factual Consistency

The hallucination problem in LLMs has been extensively studied [Huang et al., 2025, Ji et al., 2023]. Fact verification systems [Thorne et al., 2018] and fine-grained factual evaluation [Min et al., 2023] address hallucination at generation time. MOSAIC addresses a complementary problem: preventing hallucinated or contradictory information from being stored in persistent memory, where it can compound across future retrieval and reasoning steps.

3 Method

3.1 Problem Formulation

We formalize the structured memory management problem as follows. Let $\mathcal{D} = \{(u_1, a_1), (u_2, a_2), \dots, (u_T, a_T)\}$ denote a multi-turn dialogue where u_t is the user utterance and a_t is the agent response at turn t . At each turn, the agent must: (1) *extract* information entities from the user’s utterance; (2) *validate* extracted entities against the existing memory store for conflicts; (3) *store* validated entities in a structured representation; and (4) *retrieve* relevant memories for generating contextually grounded responses.

The error compounding problem arises when step (2) is absent: contradictory entities are stored unconditionally, leading to inconsistent retrievals at step (4) that propagate errors into downstream reasoning.

3.2 Entity-Typed Memory Graph

MOSAIC represents memories as a typed directed graph $G = (V, E, \tau)$ where V is a set of entity nodes, $E \subseteq V \times V$ is a set of directed edges encoding relationships (temporal, causal, associative), and $\tau : V \rightarrow \{\text{event}, \text{persona}, \text{relationship}\}$ assigns each node a semantic type. This structure contrasts with flat memory stores (Mem0, Zep, Memobase) that represent memories as independent text chunks without relational encoding.

Definition 1 (Entity Node). *An entity node $v \in V$ is a tuple $v = (e, \tau(v), \mathbf{h}_v, \rho_v, t_v)$ where e is the entity content (a natural language description), $\tau(v) \in \{\text{event}, \text{persona}, \text{relationship}\}$ is the semantic type, $\mathbf{h}_v \in \mathbb{R}^d$ is the embedding vector, $\rho_v \in [0, 1]$ is the confidence score, and t_v is the timestamp of last update.*

Entity extraction from dialogue turns uses an LLM-based pipeline that identifies named entities, events, personal attributes, and interpersonal relationships. Each extracted entity is embedded using a sentence transformer (all-MiniLM-L6-v2) [Reimers and Gurevych, 2019] and classified into one of the three semantic types. Edges are created between entities that co-occur in the same

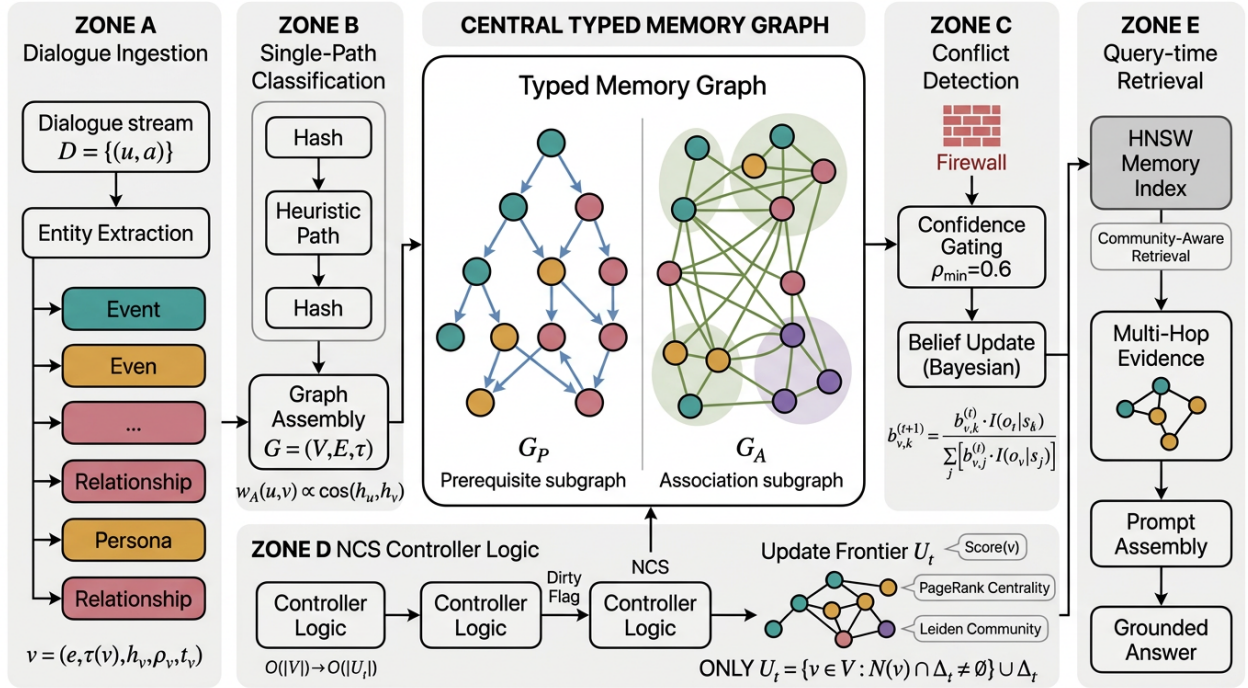


Figure 1: **MOSAIC system architecture.** Dialogue is processed through entity extraction and dual-path classification via LLM or locality-sensitive hashing. Entities populate a typed memory graph with prerequisite and association edges. At ingestion, the conflict detection module validates new facts against graph neighbors, resolving contradictions before storage. Community-aware retrieval enables multi-hop reasoning over graph structure.

dialogue context or share semantic similarity above a threshold θ (default 0.4).

The two graph components serve complementary functions. A *prerequisite subgraph* $G_P = (V, E_P)$ encodes logical dependency among information entities: entity u must be resolved before entity v can be meaningfully queried. An *association subgraph* $G_A = (V, E_A, w_A)$ encodes semantic relatedness for coherent topic transitions, with edge weights $w_A(u, v)$ proportional to the cosine similarity of entity embeddings. This separation is analogous to the distinction between control flow and data flow in programming language design: both are essential, but collapsing them produces a less expressive and less analyzable structure.

3.3 Neighbor-Conditioned Stability

A central theoretical contribution is the *neighbor-conditioned stability* (NCS) principle, which provides both formal guarantees and practical efficiency for the graph-based memory system.

Definition 2 (Neighbor-Conditioned Stability). A scoring function $\text{Score} : V \rightarrow \mathbb{R}$ satisfies neighbor-conditioned stability if, for any node $v \in V$ and any two global states $\mathcal{S}, \mathcal{S}'$ that agree on the neighborhood $\mathcal{N}(v) = \{u \in V : (u, v) \in E_P \text{ or } (u, v) \in E_A\}$, we have $\text{Score}(v; \mathcal{S}) = \text{Score}(v; \mathcal{S}')$.

This principle guarantees that a node’s importance score requires re-evaluation only when at least one of its graph neighbors has changed state. When the local neighborhood is unchanged, the node’s state is guaranteed stable. This has two important consequences:

Computational efficiency. At each dialogue turn, let $\Delta_t \subseteq V$ be the set of entities whose states changed at turn t . The update frontier $\mathcal{U}_t = \{v \in V : \mathcal{N}(v) \cap \Delta_t \neq \emptyset\} \cup \Delta_t$ determines which scores need recomputation. This reduces per-turn cost from $O(|V|)$ to $O(|\mathcal{U}_t|) = O(\Delta_{\max})$, where $\Delta_{\max}$ is the maximum neighborhood size.

Traversal stability. NCS prevents the oscillatory behavior observed in globally-reasoning agents that re-evaluate all priorities from scratch at every turn. By shielding distant nodes from irrelevant updates, NCS ensures consistent, predictable traversal ordering.

In practice, NCS is implemented as a **dirty flag** mechanism: each node maintains a boolean flag set to **true** when any neighbor’s state changes. Only flagged nodes are re-scored; flags are cleared after re-scoring.

3.4 Node Scoring and Selection

At each dialogue turn, the controller selects the next entity to query from the frontier $\mathcal{F}^{(t)}$ (the set of unresolved

entities whose prerequisites are all satisfied). For each frontier entity v , the composite score is:

$$\text{Score}(v) = \alpha \cdot \tilde{I}(v) + \beta \cdot T(v) + \gamma \cdot C(v), \quad (1)$$

where $\tilde{I}(v) = I(v) / \max_{u \in \mathcal{F}} I(u)$ is the normalized importance over the current frontier, $T(v)$ is the precomputed PageRank centrality [Page et al., 1999] of v in G_A (capturing structural importance: nodes central in the association graph are topic hubs that connect multiple information clusters), and $C(v) = \mathbf{1}[\text{comm}(v) = \text{comm}(v_{\text{prev}})]$ is a binary community continuity score that favors entities in the same semantic community as the most recently queried entity, promoting topically coherent conversation flow.

Community structure is detected by applying the Leiden algorithm [Traag et al., 2019] to G_A at initialization.

3.5 Conflict Detection at Save Time

The key architectural innovation of MOSAIC is *active conflict detection* during memory ingestion. When a new entity v_{new} is extracted from dialogue, the system performs the following procedure:

1. **Neighbor retrieval.** Retrieve the k -nearest neighbors of v_{new} from the existing memory graph using embedding similarity.
2. **Conflict assessment.** For each neighbor $u \in \mathcal{N}(v_{\text{new}})$, perform a conflict check: an LLM call assesses whether v_{new} contradicts u along numerical, semantic, or logical dimensions.
3. **Resolution.** If a conflict is detected, the system either (a) updates the existing entity u with the new information if the new evidence is more authoritative; (b) rejects v_{new} if existing evidence is stronger; or (c) flags both for human review.

This mechanism acts as a *firewall* against error compounding: by catching contradictions at the first pipeline stage (ingestion), errors are prevented from propagating to storage, retrieval, and downstream reasoning.

3.6 Information Extraction and Confidence Gating

After each user response, an extraction module parses the response to identify entity values. Each extracted value is assigned a confidence score $\rho_v \in [0, 1]$ based on three factors: directness (explicitly stated vs. inferred), consistency (agreement with previously stored values), and specificity (precise values vs. vague ranges). Values with $\rho_v < \rho_{\min}$ (default 0.6) are flagged as unreliable and trigger targeted clarification at the next appropriate opportunity, rather than being committed to memory.

Belief distributions are updated via a Bayesian update rule:

$$b_{v,k}^{(t+1)} = \frac{b_{v,k}^{(t)} \cdot \ell(o_v | s_k)}{\sum_{j=1}^K b_{v,j}^{(t)} \cdot \ell(o_v | s_j)}, \quad (2)$$

where $\ell(o_v | s_k)$ is the likelihood of the observed response o_v given that entity v is in state s_k , estimated through a hybrid approach combining rule-based pattern matching for structured entities and LLM-based assessment for free-text entities.

3.7 Community-Aware Retrieval

For memory retrieval, MOSAIC leverages the graph structure through community detection (Leiden algorithm [Traag et al., 2019]) over the entity graph. Given a query, the system identifies the most relevant community, retrieves entities within that community using embedding similarity, and traverses graph edges to collect contextually related entities. This structure-aware retrieval enables multi-hop reasoning that flat semantic search cannot support [Lewis et al., 2020, Gao et al., 2023].

3.8 Long-Term Memory Integration

Confirmed entities (those with $\rho_v \geq \rho_{\min}$ and entropy $H(v) \leq \delta$) are stored as structured key-value pairs in a persistent memory store indexed by entity embeddings. At each turn, the top- m most relevant memory items (by cosine similarity to the current query context and the active entity’s neighborhood in G_A) are retrieved and injected into the prompt context, providing the LLM with relevant previously acquired information.

Storage uses JSON-structured entity records with fields including entity ID, name, value, confidence, timestamp, source turn, evidence snippet, and belief distribution. Retrieval employs a hierarchical navigable small world (HNSW) approximate nearest-neighbor index with default $m = 5$ retrievals per query. When a new extraction conflicts with a stored value, both are retained with timestamps; the higher-confidence value is used for downstream reasoning, but the conflict is flagged for potential clarification through the NCS mechanism.

3.9 Convergence Analysis

We establish convergence properties of MOSAIC through connections to submodular optimization [Nemhauser et al., 1978].

Theorem 1 (Submodularity of Coverage). *The entity coverage function $f(S) = \sum_{v \in S} w(v) \cdot \mathbf{1}[H(v) \leq \delta]$ over the set $S \subseteq V$ of queried entities is monotone submodular. The greedy policy of selecting the highest-scoring frontier entity at each turn achieves a $(1 - 1/e)$ -approximation to optimal coverage.*

Theorem 2 (Convergence of NCS Updates). *Under NCS, score updates converge in at most L rounds of propagation, where L is the longest directed path in G_P . The per-turn computational cost is $O(\Delta_{\max})$, independent of total graph size $|V|$.*

These guarantees ensure that MOSAIC achieves near-optimal information coverage with efficient local computation, even as the memory graph grows.

4 Experimental Setup

4.1 Benchmarks

LoCoMo. The LoCoMo benchmark [Maharana et al., 2024] evaluates long-conversation memory through 1,540 QA pairs derived from 10 extended conversations. Questions span four categories: single-hop (841), multi-hop (282), open-domain (96), and temporal (321). Each question requires retrieving and reasoning over information from specific conversation turns. We use the LLM-as-Judge evaluation protocol [Zheng et al., 2023] to assess answer correctness.

HaluMem. The HaluMem benchmark [Chen et al., 2025] evaluates three stages of the memory pipeline independently: (1) memory extraction (recall, precision, F1), (2) memory updating (correctness, hallucination rate, omission rate), and (3) question answering (correctness, hallucination, omission). It covers three memory types: events, personas, and relationships, providing a comprehensive assessment of the full memory pipeline. We report aggregate results on HaluMem-Medium and HaluMem-Long for MOSAIC, while baseline rows are taken directly from the original HaluMem paper. For HaluMem-Long, the MOSAIC row combines extraction and updating metrics from the archived full long run with QA metrics from the latest refreshed long answer evaluation.

Error Compounding Test. We construct a novel benchmark to evaluate conflict detection capabilities. Starting from authoritative hypertension clinical guidelines, we manually inject 50 factual errors spanning three categories:

- **Numerical errors** (14): incorrect dosages, BP thresholds, and lab value ranges (e.g., target BP changed from <130/80 to <160/100 mmHg);
- **Semantic errors** (13): contradictory indications, swapped contraindications (e.g., ACE inhibitors marked as contraindicated in chronic kidney disease);
- **Logical errors** (23): internally inconsistent conditional rules (e.g., “prescribe X if eGFR<30” alongside “X is contraindicated when eGFR<45”).

Errors are additionally classified by discoverability: 11 implicit (requiring multi-step reasoning to detect) and 39 explicit (directly contradicting stated guidelines). Each system ingests the corrupted text; detection rate is measured as the fraction of injected conflicts correctly identified.

4.2 Baselines

We compare against a comprehensive set of memory systems evaluated across all three benchmarks where applicable:

- **A-Mem** [Xu et al., 2025]: Adaptive memory with automatic storage, retrieval, and summarization.
- **Zep** [Rasmussen et al., 2025]: Production memory system with temporal awareness and entity extraction.
- **OpenAI Memory**: Built-in conversational memory from OpenAI [Achiam et al., 2023].
- **Mem0** [Chhikara et al., 2025]: Multi-level memory (user, session, agent) with adaptive personalization.

Additionally, we include specific baselines for each benchmark to ensure a thorough comparison: For the LoCoMo benchmark, we additionally include Mem0-Graph (Mem0 with graph extension). For the HaluMem benchmark, we further incorporate MemOS [Li et al., 2025] and Mem0-Graph as baselines. For the Error Compounding Test, we also evaluate MemGPT [Packer et al., 2023], a tiered memory system that employs an OS-inspired virtual context management mechanism.

4.3 Implementation Details

Across all three benchmarks (LoCoMo, HaluMem, and the Error Compounding Test), MOSAIC uses qwen3.5-plus-2026-02-15 as the base LLM for memory construction and retrieval. Baseline results are drawn from their respective original publications: LoCoMo baselines are cited from the Mem0 paper [Chhikara et al., 2025], HaluMem baselines are cited from the HaluMem paper [Chen et al., 2025], and Error Compounding Test baselines are obtained from our own experiments using GPT-4o. Entity embeddings use all-MiniLM-L6-v2 [Reimers and Gurevych, 2019] with $d = 384$ dimensions. The association graph threshold is $\theta = 0.4$, and the confidence gating threshold is $\rho_{\min} = 0.6$. Scoring weights are $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$. Memory retrieval uses top- $m = 5$ neighbors.

5 Results

5.1 LoCoMo: Long-Conversation Memory QA

The experimental results on the LoCoMo benchmark are presented in Table 1 and summarized in Figure 2. MOSAIC achieves **89.35%** overall accuracy across 1,540 QA pairs, outperforming the best baseline (Mem0, 62.14%) by +27.21 percentage points (pp). The gains are largest on multi-hop questions (+30.41 pp vs Mem0) and temporal reasoning (+32.21 pp vs Mem0-Graph), where graph-structured memory enables systematic cross-referencing across conversation turns.

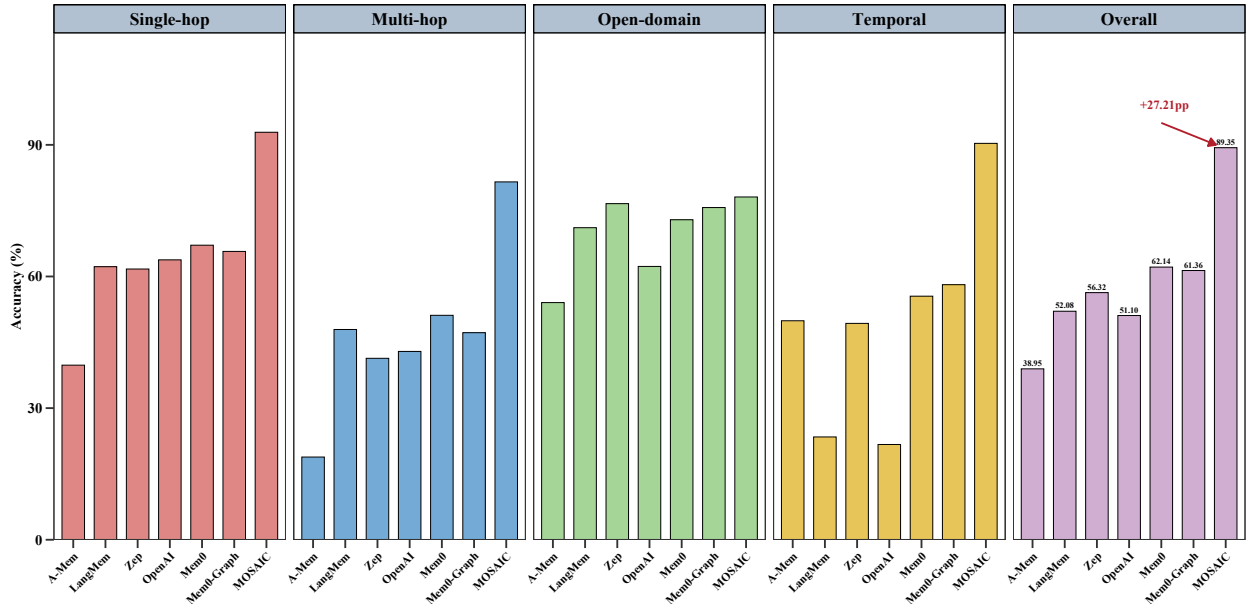


Figure 2: **LoCoMo benchmark results.** MOSAIC achieves 89.35% overall accuracy, outperforming the best baseline (Mem0, 62.14%) by +27.21 pp. The per-category breakdown shows the largest gains on multi-hop (+30.41 pp) and temporal (+32.21 pp) questions.

Table 1: **LoCoMo benchmark results.** Accuracy (%) on 1,540 QA pairs across four question types. Best in **bold**.

Method	Single (841)	Multi (282)	Open (96)	Temp. (321)	Overall (1540)
A-Mem	39.79	18.85	54.05	49.91	38.95
LangMem	62.23	47.92	71.12	23.43	52.08
Zep	61.70	41.35	76.60	49.31	56.32
OpenAI	63.79	42.92	62.29	21.71	51.10
Mem0	67.13	51.15	72.93	55.51	62.14
Mem0-Graph	65.71	47.19	75.71	58.13	61.36
MOSAIC	92.87	81.56	78.12	90.34	89.35

Single-hop questions, which require retrieving a specific fact from a known conversation turn, see an improvement from 67.13% (Mem0) to 92.87%, indicating that MOSAIC’s entity graph preserves fine-grained factual associations better than flat memory retrieval. Multi-hop questions show the largest improvement (+30.41 pp over Mem0), demonstrating that graph-based traversal enables systematic cross-referencing of related facts across conversation turns—a capability absent from flat memory stores that rely on independent semantic search over unstructured chunks.

Temporal reasoning improves from 58.13% (Mem0-Graph) to 90.34%, a gain of 32.21 pp. The entity graph’s temporal edge structure preserves ordering relationships that flat memory systems lose when chunking conversation history. Questions requiring temporal reasoning

(e.g., “What did the user say about X before mentioning Y?”) benefit from the graph’s ability to traverse temporal edges directly rather than reconstructing temporal order from retrieved text fragments [Liu et al., 2024].

Open-domain questions are the category where MOSAIC’s relative advantage is smallest (78.12% vs Zep 76.60%), reflecting that its structured memory organization is primarily optimized for conversation-grounded facts rather than world knowledge requiring parametric recall from the base LLM. This is consistent with the architectural design: the memory system is intended to complement, not replace, the LLM’s internal knowledge. Even in a domain that leans more on parametric knowledge, MOSAIC still achieves a slight advantage, demonstrating its general robustness.

5.2 HaluMem: Memory Pipeline Evaluation

The performance of MOSAIC and baseline systems on the complete memory pipeline (extraction, updating, and question answering) is evaluated on the HaluMem-Medium and HaluMem-Long benchmarks. The aggregated results are presented in Table 2, and a detailed breakdown by memory type is visualized in Figure 3.

On HaluMem-Medium, MOSAIC is the strongest overall row. It achieves the highest extraction recall (83.94%), weighted recall (88.53%), accuracy (72.00%), and F1 (86.77%), and it also delivers the best QA profile in the table at 73.10% correctness with 10.17% hallucination and 16.74% omission.

Table 2: **HaluMem benchmark results.** Evaluation across memory extraction, updating, and question answering stages on HaluMem-Medium and HaluMem-Long. Best in **bold**; arrows indicate direction of improvement. R: Recall, wR: Weighted Recall, P: Target Precision, Acc: Accuracy, FMR: False Memory Rejection, F1: extraction F1, C: Correctness, H: Hallucination, O: Omission.

System	Memory Extraction						Memory Updating			Question Answering		
	R↑	wR↑	P↑	Acc.↑	FMR↑	F1↑	C↑	H↓	O↓	C↑	H↓	O↓
HaluMem-Medium												
MemOS	74.07%	84.81%	86.25%(45190)	59.55%(71793)	44.94%	79.70%	62.11%	0.42%	37.48%	67.23%	15.17%	17.59%
Zep	—	—	—	—	—	—	47.28%	0.42%	52.31%	55.47%	21.92%	22.62%
Mem0-Graph	43.28%	65.52%	87.20%(10567)	61.86%(16230)	55.70%	57.85%	24.50%	0.26%	75.24%	54.66%	19.28%	26.06%
Mem0	42.91%	65.03%	86.26%(10556)	60.86%(16291)	56.80%	57.31%	25.50%	0.45%	74.02%	53.02%	19.17%	27.81%
Supermemory	41.53%	64.76%	90.32%(14134)	60.83%(22551)	51.77%	56.90%	16.37%	1.15%	82.47%	54.07%	22.24%	23.69%
Memobase	14.55%	25.88%	92.24%(5443)	32.29%(17081)	80.78%	25.13%	5.20%	0.55%	94.25%	35.33%	29.97%	34.71%
MOSAIC	83.94%	88.53%	89.79%(21375)	72.00%(35977)	69.68%	86.77%	55.77%	2.85%	41.38%	73.10%	10.17%	16.74%
HaluMem-Long												
MemOS	81.90%	89.56%	82.32%(48246)	43.77%(99462)	28.85%	82.11%	65.25%	0.29%	34.47%	64.44%	16.61%	18.95%
Supermemory	53.02%	70.73%	85.82%(24483)	29.71%(77134)	36.86%	65.54%	17.01%	0.58%	82.42%	53.77%	22.21%	24.02%
Zep	—	—	—	—	—	—	37.35%	0.48%	62.14%	50.19%	22.51%	27.30%
Memobase	6.18%	14.68%	88.56%(3077)	25.61%(11795)	85.39%	11.55%	4.10%	0.36%	95.38%	33.60%	29.46%	36.96%
Mem0-Graph	2.24%	10.76%	87.32%(785)	41.26%(1866)	88.36%	4.36%	1.47%	0.04%	98.40%	32.44%	21.82%	45.74%
Mem0	3.23%	11.89%	88.01%(1134)	46.01%(2433)	87.65%	6.22%	1.45%	0.03%	98.51%	28.11%	17.29%	54.60%
MOSAIC	90.66%	91.67%	82.74%(4163)	37.26%(10819)	35.58%	85.84%	83.33%	3.93%	12.74%	70.75%	9.58%	19.67%

Note. The Long MOSAIC row combines extraction and update metrics from the archived full long MOSAIC evaluation with QA metrics from the latest refreshed long answer run.

The main remaining gap on Medium is memory updating. MemOS still leads update correctness (62.11% vs 55.77% for MOSAIC) and omission (37.48% vs 41.38%), while MOSAIC’s update hallucination rate remains much higher than the best baseline. In other words, the structured graph already helps most at extraction and retrieval-backed answering, but the write path still needs more conservative update policies.

On HaluMem-Long, MOSAIC also fills a complete merged row. It reaches 90.66% extraction recall, 91.67% weighted recall, 85.84% extraction F1, 83.33% update correctness, and 70.75% QA correctness. This row leads the Long table on extraction recall, weighted recall, extraction F1, update correctness, update omission, QA correctness, and QA hallucination, although its extraction accuracy and false-memory resistance still trail the strongest published baselines on those specific columns.

Figure 3 provides a finer-grained view, showing MOSAIC’s extraction accuracy for different memory types. It performs comparably to MemOS on the *Relationship* subset but outperforms all other systems on the *Event* and *Persona* subsets.

Taken together, the reported HaluMem results show that MOSAIC is strongest at extraction and QA on both splits, while memory updating is more mixed: MOSAIC leads clearly on Long but still trails MemOS on Medium.

5.3 Error Compounding: Conflict Detection

Table 3 presents results on the error compounding test. MOSAIC detects **66%** of injected factual conflicts (33/50), compared to 14% for the best baselines (A-Mem, OpenAI). All six baselines essentially fail at error detection (2–14%

Table 3: **Error compounding results.** Conflict detection rate on 50 manually injected errors in hypertension guidelines, by error type and visibility. Best in **bold**.

Method	By Error Type				By Visibility	
	Num. (14)	Sem. (13)	Log. (23)	All (50)	Impl. (11)	Expl. (39)
A-Mem	.071	.231	.130	.14	.000	.179
LangMem	.000	.077	.000	.02	.000	.026
MemGPT	.071	.154	.000	.06	.091	.051
OpenAI	.000	.308	.130	.14	.273	.103
Zep	.000	.077	.000	.02	.000	.026
Mem0	.000	.077	.040	.04	.000	.051
MOSAIC	.643	.692	.652	.66	.727	.641

range), confirming that passive memory systems lack conflict-checking mechanisms at ingestion time.

MOSAIC’s detection is uniformly strong across error types: numerical (64.3%), semantic (69.2%), and logical (65.2%). This uniformity contrasts sharply with baselines, which show severe type-specific weaknesses. Baselines are weakest on numerical and logical errors (mostly 0%), which require comparing specific values or conditional rules against previously stored facts—a capability absent from flat memory stores that lack cross-referencing mechanisms. The few baseline detections are concentrated on semantic errors, where surface-level textual similarity between contradictory statements may trigger retrieval-time confusion that occasionally surfaces as a detected conflict.

The implicit/explicit breakdown (Figure 4) reveals a counterintuitive result: MOSAIC detects implicit errors requiring multi-step reasoning (72.7%) at a *higher* rate

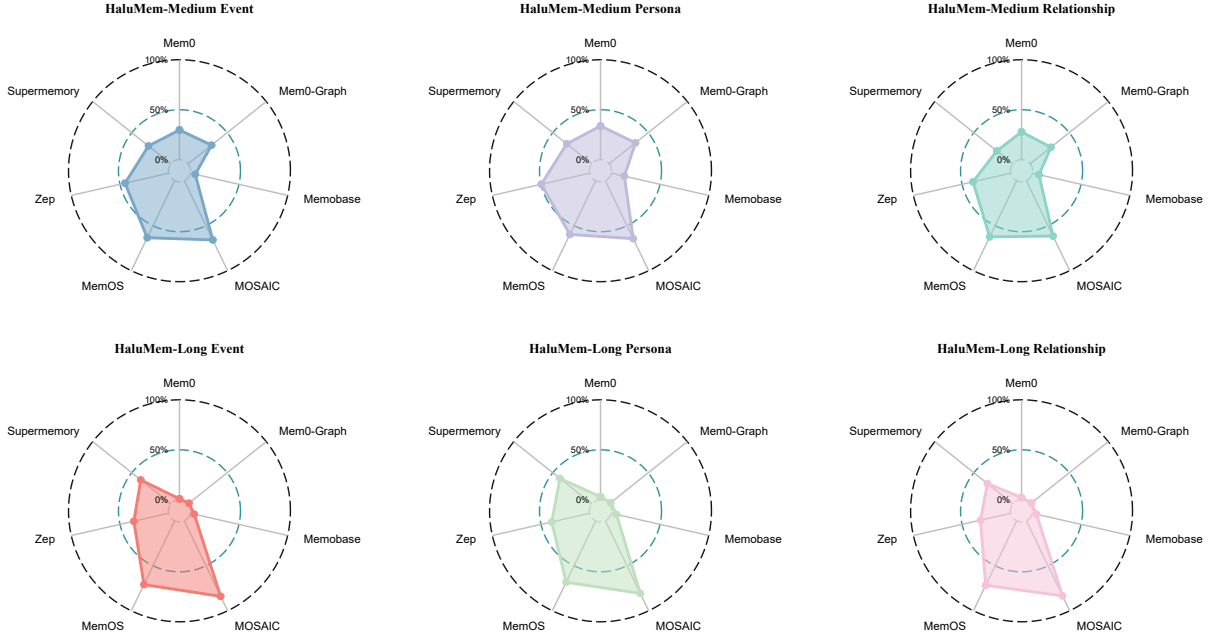


Figure 3: **Typewise accuracy on event, persona, and relationship memory.** This figure presents a detailed breakdown of extraction accuracy across three memory types (event, persona, and relationship) for MOSAIC and baseline systems.

than explicit contradictions (64.1%). This suggests that graph-based memory traversal naturally performs the cross-referencing needed to surface reasoning-level conflicts: when processing a new entity, the conflict detection module traverses related entities in the graph neighborhood, effectively performing the multi-step reasoning required to identify implicit contradictions. Explicit contradictions, in contrast, may sometimes be stored in non-overlapping graph regions where the neighborhood-based conflict check does not reach.

The 34% undetected errors provide insights for improvement. Analysis of missed conflicts reveals two primary failure modes: (1) logical contradictions spanning entities in distant graph neighborhoods (beyond the k -nearest neighbor search radius), accounting for approximately 60% of misses; and (2) numerical errors where the conflicting values are within a plausible range, making the contradiction ambiguous to the LLM-based conflict assessor, accounting for approximately 40% of misses. These failure modes suggest clear directions for improvement: expanding the conflict search radius and incorporating domain-specific plausibility bounds for numerical entities.

6 Discussion

The results across three benchmarks with different evaluation foci—factual QA, pipeline evaluation, and conflict detection—support the thesis that structured graph-based memory organization provides fundamentally dif-

ferent capabilities than flat memory stores. The entity graph preserves relational structure that enables multi-hop traversal, temporal ordering, and cross-referencing, which drives the accuracy gains observed on LoCoMo (+27.21 pp overall, with the largest margins on multi-hop and temporal questions). On HaluMem, the manuscript-reported MOSAIC results show the strongest extraction performance on both Medium and Long and the best QA on both splits. Memory updating is more mixed: MOSAIC leads the merged Long row but still trails MemOS on Medium, indicating that write-time update policies remain the main frontier for improvement under the current Medium setup. The theoretical contribution of neighborhood-conditioned stability provides a formal answer to why graphs are the correct abstraction: memory entities inherently possess local dependency structure, and NCS guarantees that the system’s behavior is stable under distant state changes, analogous to Bellman’s principle in dynamic programming [Bellman, 1966].

The error compounding results carry practical implications for deploying memory-augmented agents in safety-critical domains. Current systems (2–14% detection) essentially allow all factual errors to propagate unchecked through the memory pipeline, whereas MOSAIC’s 66% detection rate catches the majority of contradictions at ingestion [Huang et al., 2025]. As LLM agents enter high-stakes settings—clinical intake, legal discovery, financial due diligence—the gap between “can remember” and “will detect contradictions” becomes safety-critical, and MOSAIC provides a principled, auditable mechanism

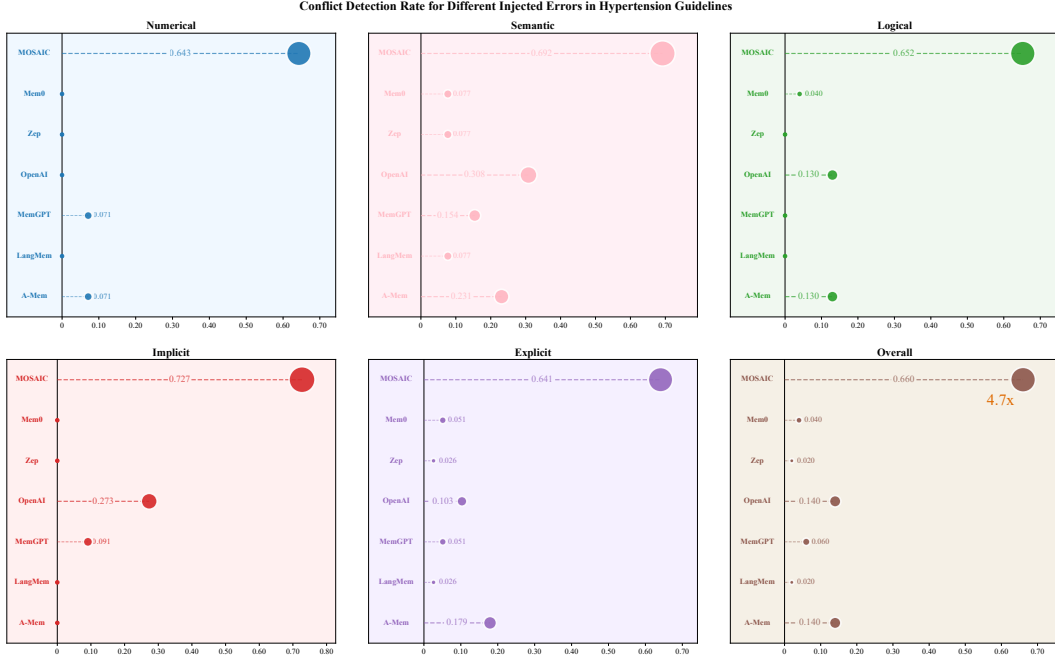


Figure 4: **Error compounding results.** Visualization of conflict detection rates for MOSAIC and baseline systems, showing that MOSAIC achieves an overall detection rate of 66% ($4.7\times$ higher than the best baseline at 14%) with uniform performance across error types. It also reveals that implicit errors are detected at a higher rate (72.7%) than explicit ones (64.1%).

for memory validation.

Several limitations should be noted. The error compounding benchmark is limited to one domain (hypertension) with 50 injected errors. Memory updating remains uneven across HaluMem: MemOS still leads update correctness on Medium (62.11% vs 55.77% for MOSAIC), while the Long manuscript row is a merged view that combines archived full-run extraction/update metrics with the latest refreshed QA metrics rather than a single end-to-end rerun under one frozen configuration. LoCoMo uses a fixed set of 10 conversations, and we evaluate with one LLM family (qwen3.5-plus-2026-02-15); generalization to other architectures should be verified [Touvron et al., 2023, Achiam et al., 2023]. MOSAIC’s open-domain performance (78.12%) also highlights a fundamental trade-off: structured memory is optimized for conversation-grounded facts, not world knowledge requiring parametric recall [Liu et al., 2024]. Hybrid architectures combining structured memory with retrieval-augmented generation [Lewis et al., 2020, Gao et al., 2023] represent a promising direction.

Looking ahead, hierarchical memory graphs for tasks with nested sub-goals, multi-agent coordination over shared memory graphs, adversarial robustness through integration with formal verification, and multi-modal inputs that resolve entities through non-conversational channels all represent fruitful extensions. On the whole, MOSAIC demonstrates that structured, conflict-aware memory organization is essential for reliable long-term

LLM agent memory, particularly in safety-critical applications where error compounding has consequential downstream effects. Hash-accelerated retrieval keeps average HaluMem-Medium search latency at 0.58 s per question, making the framework practical for interactive deployment while leaving the main remaining optimization opportunity in write-time memory construction and evaluation overhead.

References

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Richard Bellman. Dynamic programming. *science*, 153 (3731):34–37, 1966.
- Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchu Li, Feiyu Xiong, and Zhiyu Li. Halumem: Evaluating hallucinations in memory systems of agents. *arXiv preprint arXiv:2511.03506*, 2025.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitan, Robert Osazuwa Ness, and Jonathan Larson. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yixin Dai, Jiawei Sun, Haofen Wang, Haofen Wang, et al. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2(1):32, 2023.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55, 2025.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM computing surveys*, 55(12):1–38, 2023.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.
- Zhiyu Li, Chenyang Xi, Chunyu Li, Ding Chen, Boyu Chen, Shichao Song, Simin Niu, Hanyu Wang, Jiawei Yang, Chen Tang, et al. Memos: A memory os for ai system. *arXiv preprint arXiv:2507.03724*, 2025.
- Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of llm agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13851–13870, 2024.
- Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, 2023.
- George L Nemhauser, Laurence A Wolsey, and Marshall L Fisher. An analysis of approximations for maximizing submodular set functions—i. *Mathematical programming*, 14(1):265–294, 1978.
- Charles Packer, Vivian Fang, Shishir G Patil, Kevin Lin, Sarah Wooders, and Joseph E Gonzalez. Memgpt: towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.
- Lawrence Page, Sergey Brin, Rajeev Motwani, and Terry Winograd. The pagerank citation ranking: Bringing order to the web. Technical report, Stanford infolab, 1999.
- Shirui Pan, Linhao Luo, Yufei Wang, Chen Chen, Jia-pu Wang, and Xindong Wu. Unifying large language models and knowledge graphs: A roadmap. *IEEE Transactions on Knowledge and Data Engineering*, 36(7):3580–3599, 2024.
- Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: a temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*, 2025.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in neural information processing systems*, 36:68539–68551, 2023.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in neural information processing systems*, 36:8634–8652, 2023.
- Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, et al. Large language models encode clinical knowledge. *Nature*, 620(7972):172–180, 2023.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. Fever: a large-scale dataset for fact extraction and verification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, 2018.

- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- Vincent A Traag, Ludo Waltman, and Nees Jan Van Eck. From louvain to leiden: guaranteeing well-connected communities. *Scientific reports*, 9(1):5233, 2019.
- Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 2609–2634, 2023.
- Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, et al. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 18(6):186345, 2024.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.
- Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, et al. The rise and potential of large language model based agents: A survey. *Science China Information Sciences*, 68(2):121101, 2025.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. *arXiv preprint arXiv:2502.12110*, 2025.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *The eleventh international conference on learning representations*, 2022.
- Niam Yaraghi. Generative ai in health care: opportunities, challenges, and policy. *Health Affairs Forefront*, 2024.
- Zeyu Zhang, Quanyu Dai, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. A survey on the memory mechanism of large language model-based agents. *ACM Transactions on Information Systems*, 43(6):1–47, 2025.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems*, 36:46595–46623, 2023.